came in the form of particle-like quanta. Meanwhile, the fact that atoms are stable and the observation of how light was emitted from gases inspired Bohr to suggest that electrons could only move along certain allowed orbits, with occasional jumps between them. Heisenberg, Born, and Jordan elaborated this story of probabilistic jumps into a full theory, matrix mechanics. From another angle, de Broglie pointed out that if we think of matter particles such as electrons as actually being waves, we can derive Bohr's quantized orbits rather than postulating them. Schrödinger developed this suggestion into a full-blown quantum theory of its own, and it was ultimately demonstrated that wave mechanics and matrix mechanics were equivalent ways of saying the same thing. Despite initial hopes that wave mechanics could explain away the apparent need for probabilities as a fundamental part of the theory, Born showed that the right way to think about Schrödinger's wave function was as something that you square to get the probability of a measurement outcome.

Whew. That's quite a journey, taken in a remarkably short period of time, from Planck's observation in 1900 to the Solvay Conference in 1927, when the new quantum mechanics was fleshed out once and for all. It's to the enormous credit of the physicists of the early twentieth century that they were willing to face up to the demands of the experimental data, and in doing so to completely upend the fantastically successful Newtonian view of the classical world.

They were less successful, however, at coming to grips with the implications of what they had wrought.

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* Annoyingly, the electron accelerates in precisely the opposite direction that the electric field points, because by human convention we've decided to call the charge on the electron "negative" and that on a proton "positive." For that we can blame Benjamin Franklin in the eighteenth century. He didn't know about electrons and protons, but he did figure out there was a unified concept called "electric charge." When he went to arbitrarily label which substances were positively charged and which were negatively charged, he had to choose something, and the label he picked for positive charge corresponds to what we would now call "having fewer electrons than it should." So be it.

* I've emphasized that there is only one wave function, the wave function of the universe, but the alert reader will notice that I often talk about "the wave function of a particle." This latter construction is perfectly okay if—and only if—the particle is unentangled from the rest of the universe. Happily, that is often the case, but in general we have to keep our wits about us.